Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Non-Intrusive Assessment of Motion Sickness of Occupants in Automated Vehicles: A Real-time Deep Learning Framework Integrating Facial and Behavioral Features**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical and Materials		Mechanical and Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Automobile / Rail Vehicles	Safety Improvement Technologies	Automobile / Rail Vehicles	Vehicle Intelligence Technologies
Concept of Proposed Technology	**Problem & significance** Motion sickness is a persistent barrier to passenger acceptance of automated vehicles (AVs). As automation increases, all occupants become passengers and are more likely to read or view screens, exacerbating sensory conflict (the mismatch between sensed and expected motion). In current manually driven vehicles, estimates suggest that already 30-65% of passengers experience carsickness at least some times. This number is expected to rise much further in automated scenarios where everyone is a passenger [1]. Additionally, given that there is no driver present within the vehicle, there need to be ways of ensuring that the passengers are still healthy to ensure safe and comfortable operations for all occupants. This			

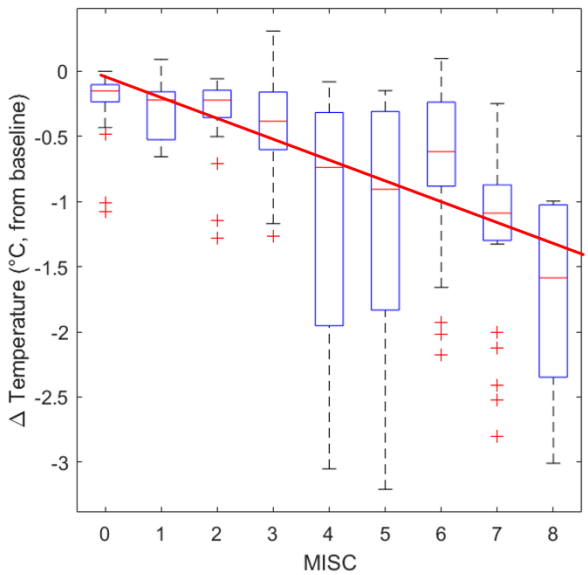
is especially relevant in the case where occupants may be required to take over control of the vehicle when automation reaches its operational limits. Given that the thresholds for motion sickness/nausea are highly dependent on the specific person, this requires active monitoring of the occupants of the vehicle.

The first step towards overcoming both these challenges is to be able to detect when the occupants of the vehicle are feeling nauseous or motion sick through non-contact measurement. This would allow for non-invasive measurement of the state of the occupants, such that the vehicle can ensure safe operation, or modify its behaviour to optimize for comfort (instead of time/distance, as standard), where needed.

Proposed solution

This project aims to leverage advances in in-cabin sensing technologies as well as edge-computing based AI to identify when an occupant of the vehicle is falling ill.

In a previous project, TNO explored the use of a Deep-Infrared Camera to detect the nausea developed in a participant during simulator experiments. This was proven to be an effective measure, with a marked reduction in the forehead temperature with higher reported motion sickness (measured through the subjective MISC scale), acting as a direct indicator of motion sickness [2], [3]. This forms the foundation of this project.



Change in forehead skin temperature measured at different levels of motion sickness based on the MISC scale in 14 participants during vertical off-axis rotation.

One important disadvantage of thermal imaging is its cost. This may be an important obstacle towards commercialization in vehicles. Therefore, this project aims to explore the use of more commonly available and lower cost Visual (RGB) cameras, as well as Near-Infrared (NIR) cameras for the detection of motion sickness in (automated) vehicles and to compare performance to that with deep infrared, thermal cameras.

This project will:

1. Form a ground-truth dataset, by means of simulator-based experimental studies in TNO's dedicated motion simulators, outfitted with RGB, NIR and thermal cameras as well as electrocardiogram (ECG), skin conductance (EDA) and respiration measures. Motion sickness will be measured during motion exposure using the MISC subjective measurement scale (developed by TNO [4], [5]), at 1 min intervals.
2. Define the correlation between the observed physiological and image-based parameters and the subjective motion sickness state.

	3. Develop and prototype these sensors. **Intended application** • Highly automated (SAE L3 or higher) passenger or commercial vehicles. • Manually driven vehicles, such as taxi-cabs, where the driver can modify their driving behavior based on the state of the passenger. • Other markets: This also enables applications in the marine/rail/VR domains (provides additional future scope and possibility for broader valorisation of knowledge through methodology transfer).

Rationale for Joint R&D Project	**<u>Research gap to be addressed</u>**
	
	• Identify which biomarkers and physiological/body posture/facial expressions can provide an indication of the development of discomfort, nausea or motion sickness.
	• Develop a pipeline that utilizes AI and in-cabin sensors to detect such developments.
	
	<u>Complementarity</u>
	
	• **TNO:**
	  ○ Over 30 years of prior experience in motion sickness & human factors research, across the automotive, defence, space, cyber, maritime and other sectors.
	  ○ TNO has specialized research facilities that can facilitate this research, including the Desdemona (6-DoF Motion Simulator), Limosine (40 m linear simulator), hexapod driving simulator, rotation chair, and TNO Carlabs (automated test vehicles).
	• **Korean SME(s):**
	  ○ Expertise in the development of in-vehicle cabin sensor technologies as well as AI pipelines.
	  ○ Productization and commercialization expertise.
	
	<u>Why international collaboration</u>
	
	Mitigation of motion sickness requires interdisciplinary co-development (vehicle dynamics + human factors + physiology) and multi-demographic validation.
	
	AV supply chains are inherently global. The opportunity to co-design the solution with Korean SMEs (working on in-cabin sensing hardware and detection pipelines) accelerates the possibility to scale-up the innovation from TNO, towards application in the automotive industry.
	
	<u>Outcome</u>
	
	Prototype of in-cabin sensing visual (RGB), Near Infrared (NIR) and/or Thermal camera-based sensors that can detect the motion comfort state of the occupants. Annotated datasets that can be used for validation and development efforts for such sensors.

Key Words	1. **Occupant monitoring technology** to detect the occupants' sickness state:
	  a. Normal;
	  b. Mild discomfort e.g., headache, yawning, sweating etc.
	  c. Moderate discomfort e.g., nausea; and
	  d. Severe discomfort.
	2. **Vision-based AI** that keeps track of occupant's features including heart rate, head pose, 3D body pose, facial expression, and behavior (action)
	3. **On-device AI** to process data locally on the vehicle's ECU (Neural Processing Unit), ensuring low latency and GDPR privacy compliance (no video leaves the car).

3. Research and Development Objectives & Plan

(max 10 pages)

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)

R&D Objectives

The overarching goal of this project is to develop a non-intrusive and privacy-preserving AI system for the early detection of kinetosis (motion sickness) in vehicles (with a special emphasis on automated vehicles).

Objective 1: Establish a Multimodal "Ground Truth" Dataset

Collect and synchronize high-fidelity sensor data (RGB, NIR, Thermal) with physiological ground truth (ECG) and controlled vestibular stimuli using simulator based testing. Simultaneously, self-report ratings of motion sickness severity (MISC)will be collected at regular intervals (1 min).

Objective 2: Correlate (In)visible Biomarkers with Sickness States

Quantify the relationship between visible behaviors (yawning, head drooping, unstable body pose) and invisible autonomic responses (facial thermal shifts, heart rate variability) to classify occupants into normal and mild, moderate and severe discomfort.

Objective 3: Develop a RGB/NIR camera based motion sickness detection system

Develop AI-based algorithms that can detect the changes in aforementioned visible and invisible autonomic responses to classify occupants into normal and mild, moderate and severe discomfort.

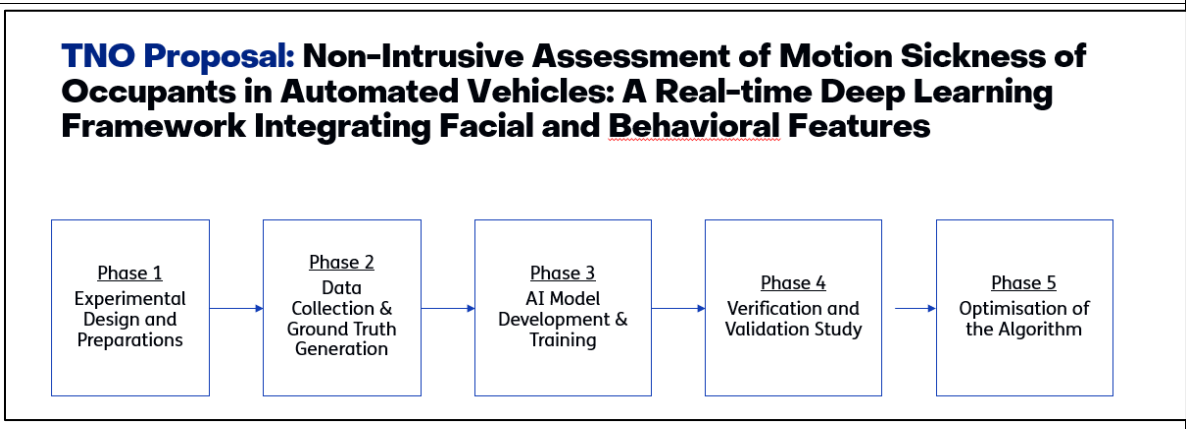
Objective 3: Implement High-Performance Edge Computing

Optimize the complex multimodal AI models to run on a vehicle's embedded NPU (Neural Processing Unit) with low processing time (<30ms) and ensuring that no biometric video data is stored or transmitted, complying with GDPR and privacy-by-design principles.

R&D Target Specifications

- Dataset covering ≥ 50 people for use in future development and validation effort..
- Detection accuracy: > 90% of people with motion sickness detected; > 80% severity classification accuracy
- Performance: Processing time of the model < 30ms.

R&D Plan



Phase 1: Experimental Design and Preparations (Months 0–6; Lead: TNO)

- Design provocation scenarios (including motion profiles) in suitable motion simulator (e.g., Off Vertical Axis Rotation, or continuous longitudinal acceleration) known to induce kinetosis.
- Secure IRB approval for human-based research.
- Define "Stop Criteria" (e.g., reaching MISC Level 8) for participant safety.

Milestone M1: The experiment design is completed, and the experimental phase can begin.

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Phase 2: Data Collection and Ground Truth Generation (Months 6–10; Lead: TNO)

- Recruit diverse group of participants (e.g., age, gender, motion sickness susceptibility).
- Administer experiments:
  - Collect RGB, NIR, and thermal image data.
  - Monitor the test equipment and the participants.
  - Reference monitoring: medical-grade sensors (e.g., a selection from ECG, EDA, respiration, EGG).
  - Continuous MISC scoring every 60 seconds.
- Data processing: correlate physiological data with MISC ratings; privacy anonymization.

Milestone M2: Experimentation completed, dataset delivered to SME(s).

Phase 3: AI Model Development and Training (Months 10–18; Lead: SME)

- Biomarker extraction: algorithms for facial thermal, visual (actions, yawning), physiological (rPPG HR) detection using the equipped sensors.
- Benchmarking biomarker based motion sickness estimates against MISC ratings collected during the experiment (TNO).
- Benchmarking image-based biomarker estimates with physiological measurements collected during the experiment (TNO).
- Fusion model training: Temporal Convolutional Network (TCN) or LSTM to correlate features with MISC Score.
- Optimization: quantize model for low-power automotive NPU.

Milestone M3: MVP system completed.

Phase 4: Verification and Validation Study (Months 18–21; Lead: TNO)

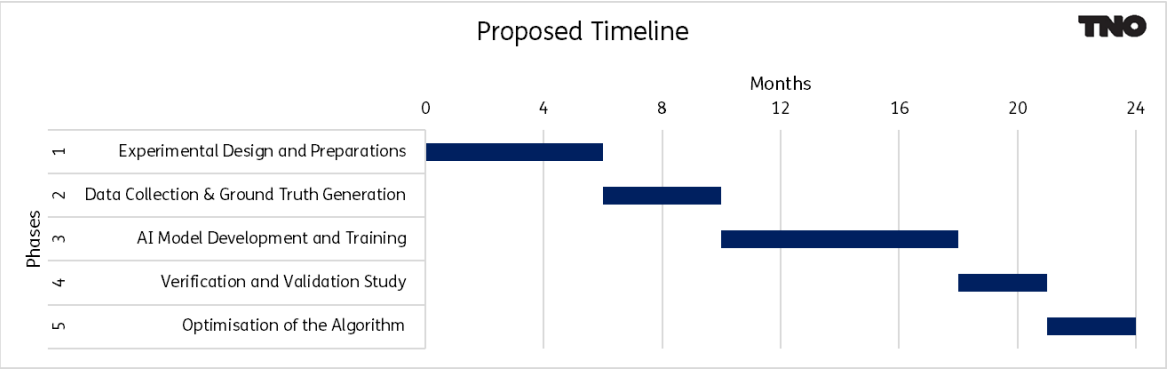
- Conduct validation trials with SME sensor via simulator study (similar structure as Phase 2, but smaller scale).
- Process data and quantify accuracy of the SME's sensor/algorithm based on observed MISC scores.

Milestone M4: Report on accuracy of the developed sensor/algorithm,.

Phase 5: Optimisation of the Algorithm (Months 21–24; Lead: SME)

- Latency minimization: processing time under 30ms.
- Initial demonstrations.

Milestone M5: Optimized algorithm/sensor.



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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	TNO provides expertise in motion sickness and human factors research, and provides access to dedicated motion sickness research facilities. TNO provides the datasets required for algorithm development and validation.
	
	Phase 1: Experimental Design & Protocol Definition
	
	• Provocation Scenarios: Design specific motion profiles known to induce kinetosis, based on TNO's extensive experience with motion sickness experiments.
	• Ethics & Safety: Secure approval from the Internal Review Board (for humans based research). This is critical as the study intentionally causes physical distress (nausea).
	• Protocol Definition: Define the "Stop Criteria" (e.g., reaching MISC Level 8) to ensure participant safety while maximizing data utility.
	• Define the motion profile to be tested.
	
	Phase 2: Data Collection & Ground Truth Generation
	
	• Participant Management: Recruit a diverse pool of participants, balancing various relevant demographic factors.
	• Conduct the experiment:
	  ○ Sensor-data collection: TNO will instrument the simulator with sensors aimed at collecting RGB, NIR and thermal image data.
	  ○ Reference Monitoring: Instrument participants with medical-grade sensors to record "invisible" ground truth: Electrocardiogram (ECG), respiration, Electrodermal Activity (EDA), Gastric Myoelectric Activity (EGG)
	  ○ Continuous MISC Scoring: Participants are prompted to provide a rating of motion sickness severity every 60 s.
	• Data Processing: Provide a synchronized dataset combining the image data, the physiological data and the MISC measurements to SME.
	• Privacy: Ensure compliance with the data governance policy.
	
	Phase 3: AI Model Development & Training
	
	• Physiological Correlation: Verify if SME's image-based heart rate and respiration rate estimates match the medical ECG and respiration data to validate the "virtual sensor" accuracy.
	
	Phase 4: Validation & Closed-Loop Testing
	
	• Validation Trials: Conduct a second round of tests with the sensor from the SME.

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	 TNO's Desdemona 6DoF simulator environment[1]
Expected R&R of Korean SME Partner	The SME is responsible for developing the vision-based AI system that translates raw camera data into a sickness score, and optimizes it for automotive ECUs, and commercializes the solution for OEMs. **Phase 1: Experimental Design & Protocol Definition** • Sensor Integration: Design the mounting rig to install RGB-IR and Thermal cameras inside the OVAR cabin without interfering with the participant's safety harness or view. • Data Spec: Define the required video resolution, frame rates (e.g., 60fps for micro-expressions), and thermal sensitivity (e.g., NETD < 50mK) needed for the AI. • Provide test sensors to TNO in the Netherlands. **Phase 3: AI Model Development & Training** • Biomarker Extraction: Develop algorithms to extract specific features from the raw video: • Thermal: Quantify temperature changes in facial areas. • Visual: Detect changes in muscle activation, facial expressions and appearance associated with discomfort and motion sickness. • Physiological: Estimate Heart Rate (rPPG) from facial color/brightness changes and respiration rate from skin temperature modulation. • Optimization: Quantize the model to run on a low-power automotive NPU (e.g., Qualcomm/NVIDIA/Ti edge chips). **Phase 4: Validation & Closed-Loop Testing** • Validation Trials: Conduct the first round of tests on validation/test dataset to check if the model predicts "Sickness Score". **Phase 5: Optimization of the algorithm** • Latency Minimization: Ensure the processing time is under 30ms to allow effective mitigation.

5. Expected Outcomes/Impact

[1] Desdemona: The ultimate motion simulator | TNO

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Expected Outcomes and Impact	**Technical outcomes** • Validated unobtrusive motion sickness detection system for automated vehicles. • Annotated datasets for further development and validation. • Edge AI algorithms for real-time, privacy-preserving occupant monitoring. **Industrial & economic impact** • Differentiated comfort and safety feature for OEMs. • Potential reduction in motion-sickness-related complaints, increased passenger comfort and higher passenger acceptance of automated functions.

6. References

[1] C. Diels and J. E. Bos, "Self-driving carsickness," *Applied Ergonomics*, vol. 53, pp. 374–382, Mar. 2016, doi: 10.1016/j.apergo.2015.09.009.

[2] L. Nijskens *et al.*, "An EO/IR monitoring system for noncontact physiological signal analysis in automated vehicles," in *Autonomous Systems for Security and Defence*, J. Dijk and J. L. Sanchez-Lopez, Eds., Edinburgh, United Kingdom: SPIE, Nov. 2024, pp. 1–14. doi: 10.1117/12.3033956.

[3] S. Louvenberg, J. E. Bos, L. Nijskens, S. E. van der Hurk, and J. L. Souman, "Facial Skin Temperature as a Biomarker for Motion Sickness," *Applied Ergonomics*, revision under review.

[4] J. E. Bos, S. N. MacKinnon, and A. Patterson, "Motion Sickness Symptoms in a Ship Motion Simulator: Effects of Inside, Outside, and No View," *Aviation, Space, and Environmental Medicine*, vol. 76, no. 12, Art. no. 12, 2005.

[5] A. J. C. Reuten, S. A. E. Nooij, J. E. Bos, and J. B. J. Smeets, "How feelings of unpleasantness develop during the progression of motion sickness symptoms," *Exp Brain Res*, vol. 239, no. 12, Art. no. 12, Sep. 2021, doi: 10.1007/s00221-021-06226-1.